

Seonghwan Lim

Computer Vision Engineer

Champaign, IL | +1 (312) 287-3570 | qube20101219@gmail.com | seonghwan97.github.io

SUMMARY

Computer vision engineer with an M.S. in Computer Science and experience developing, evaluating, and optimizing end-to-end computer vision and machine learning systems. My work spans data preparation, model development, performance analysis, and validation.

EDUCATION

Illinois Institute of Technology — M.S. Computer Science, May 2026, GPA 3.9/4.0

Seoul Nat'l University of Science and Technology — B.S. Electrical and Information Engineering, Aug 2023, GPA 3.6/4.0

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Bash

Computer Vision: Object detection, classification, multi-object tracking, 2D/3D pose estimation, image processing, OpenCV

Machine Learning: PyTorch, CNNs, Vision Transformers, multi-task learning, data augmentation, scikit-learn, XGBoost

Data and Tools: Dataset design, labeling QC, NumPy, Pandas, Git, Linux, Docker, real-time GPU inference (PyTorch, CUDA), embedded systems

EXPERIENCE

Research Assistant, iConSenSe Lab, Illinois Institute of Technology

Jan 2025 - Present

Camera-only worker safety monitoring; mixed-reality safety training analysis

- Built a camera-only worker monitoring system on ordinary video, no wearables. A YOLO pose model extracts keypoints, ByteTrack holds worker identity across frames, and downstream heads roll activity, work state, and posture into a 10-second exposure score at a median 23 fps for 3 to 9 workers.
- Checked every module against an independent reference across 42 subjects: 0.861 activity accuracy, 0.775 work-state F1, and 0.793 correlation with a 3D-pose reference. Scoped workload as an exposure indicator rather than a fatigue measurement after it reached only 0.526 against indirect calorimetry, cross-checked in the field against a CO2 sensing rig I built on Raspberry Pi and Jetson.
- On a separate mixed-reality study, derived movement and energy-expenditure features from single-camera video across 61 HoloLens 2 sessions using 2D pose estimation, MotionBERT 3D lifting, and spatiotemporal feature extraction, with no depth camera or motion capture rig.

Research Assistant, MMCom Lab, Illinois Institute of Technology

Jan 2025 - Jul 2026

Fine-grained road surface classifier

- Trained a classifier over 27 road-surface types on RSCD, roughly 960k training images against a 50k test split. Traced most errors to a long-tailed class distribution and to surface types that look nearly identical in a single frame.
- Fixed the data before the model, filling out rare classes with diffusion-generated images, then tried multi-task and mixture-of-experts heads to separate the look-alike classes, moving EfficientNet-B0 from 91.70% to 93.33%.

Research Assistant, Bhushan Research Group, Illinois Institute of Technology

Jan 2025 - May 2026

Image-based process control for microfluidic mixing

- Wrote the measurement code that registers and normalizes CFD contour and experimental fluorescence images, then computes a 0 to 1 mixing index from outlet pixel intensity.
- Benchmarked SVR, MLP, XGBoost, Random Forest, and linear regression as CFD surrogates. SVR won on held-out data (MAE 0.0014, R2 0.9998), close enough to replace a full CFD solve during inverse screening, confirmed with fresh CFD runs and six fluorescence experiments.

Undergraduate Researcher, VCL Lab, Seoul National University of Science and Technology

Jan 2023 - Dec 2023

Two-stage parcel inventory recognition

- Built two-stage recognition for stacked parcels: DINO and YOLO detect product regions, Weighted Boxes Fusion merges them into cleaner boxes than either alone, then EfficientNet-B7 with test-time augmentation classifies each crop.
- Placed 1st of 19 teams, identifying 56.6% of the 53 evaluation products against 49.1% for the runner-up, at 0.80 mean IoU.

SELECTED PROJECTS

Road Damage Analysis — Pipeline that pulls road geometry from OpenStreetMap and the matching Street View frames, detects pavement damage with YOLO, and renders an interactive map with clustering and a heatmap for triage.

Marionette Motion Control — Kinect RGB-D 3D joint capture driving a C++ servo control layer on an embedded controller, each servo clamped to its reachable range so a bad pose estimate cannot damage the rig.

PUBLICATIONS, AWARDS, AND PRESENTATIONS

Vision-Based Construction Worker Safety Monitoring with Pose-Derived Workload and Ergonomic-Risk Proxies

Under review, Automation in Construction

2026

Research Grant, BuiltWorlds Global Summit (\$12,500)

2026

1st Prize, CJ Logistics Future Technology Challenge (\$15,000)

2023

Sensor-Free Wearable System for Activity and Fatigue Detection in Worker Safety Monitoring, BuiltWorlds Global Summit

2026

Real-Time Control of Microfluidic Mixing via Machine Learning-Driven Prediction, BMES Annual Meeting

2025

Development of Marionette Motion Control System, ICROS-KROS Conference

2022